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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	BEBZA
Selected Domain	Domain 1 – Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 – Decent Work and Economic Growth (targets 8.2, 8.5), with SDG 4 – Quality Education (targets 4.3, 4.4)
Project / Solution Name	Future Edge
Name (Leader)	Azimkhan Bekassyl
Email & Phone (Leader)	beka2007azimut@gmail.com · +7 701 770 7593
University Affiliation(s)	XMUM
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing which jobs exist faster than training programmes can be updated. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have received formal training (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation expects more than 600,000 workers to need reskilling within three to five years. The World Economic Forum's Future of Jobs Report 2025 (Malaysia profile) projects that the share of work tasks done mainly by people will fall from 47% to 33% by 2030, and that data-entry clerk jobs will shrink by 28% while AI and machine-learning specialist jobs grow by 29%.

The people most affected are not short of information about AI; they are short of a personal, forward-looking answer. A fresh graduate choosing a first job, an admin executive whose spreadsheet work is being automated, or a final-year engineering student cannot tell which of their own skills AI will take over, which will keep them valuable, and what single thing to learn next. Existing career tools give a one-time keyword match against a static list, and "AI career coaches" are chatbots that can invent jobs or courses.
This matters economically (SDG 8 – productive employment) and socially (SDG 4 – relevant skills and lifelong learning), most of all for workers without paid career advice. The specific moment we design for is: a person deciding what to learn next, before AI makes that decision for them.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Future Edge turns a CV into a three-year career forecast, shown as a weather forecast: an Edge Score from 0 to 100 for each year, meaning "how hard you will be to replace by AI". A user goes from CV to ranked roles and an ordered learning plan in about three minutes.

Three models work together:

Model 1 – AI exposure: a text classifier trained on 19,265 human-rated O*NET work tasks (Eloundou et al., 2023) reads each skill and the user's own CV sentence and estimates how much of that work AI can do (73.4% accuracy on unseen occupations, versus 53.2% for a baseline).
Model 2 – AI-trend signal: monthly arXiv paper counts for ten AI-capability areas, forecast with a damped trend and an 80% range, decide how fast AI tools arrive for each skill.
Model 3 – Catch the AI: the user reviews an AI-written work sample with planted mistakes, measuring the judgment that keeps humans in charge of AI.
These combine with Malaysian job-market data (WEF Future of Jobs 2025, DOSM wages, TalentCorp MyCOL) into an explainable score under moderate and fast AI scenarios; the plan starts with the “One Move” that raises the score most. Users can pick SDG themes that re-rank their roles, use the app in English, Bahasa Malaysia or Chinese, and see a fairness check proving that changing name, gender, age or race does not change their result. An HR team view shows a team's skill gaps. No generative model writes recommendations, so nothing is invented. From the SDC workshops we use Replit for hosting and JSON-schema structured outputs when the optional Gemini API reads hard CVs; every phrase is checked against the CV.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Conceptual tech stack:

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS (mobile-friendly, step-by-step wizard, forecast and quest map)
Backend / API	Python 3, FastAPI, Uvicorn – endpoints for CV upload, forecast, pathway, Catch the AI and trend refresh
AI models	Model 1: TF–IDF + logistic regression (trained with scikit-learn on CPU, exported to JSON, runs in plain Python). Model 2: damped log-linear trend with 80% intervals. Model 3: exact scoring of a performance task.
Data store	Versioned JSON catalogs in the repository: 52 skills, 34 roles, 53 learning resources, SDG targets, trend snapshot. No personal data is stored.
External data	arXiv API (trend signal, refreshable), O*NET / GPTs-are-GPTs labels, WEF Future of Jobs 2025, DOSM, TalentCorp MyCOL
GenAI API (optional)	Google Gemini API with a JSON schema (structured outputs) for photo or complex-layout CVs
Cloud hosting	Replit autoscale deployment (build.sh → start.sh)

System flow (block diagram): User (browser, phone or laptop) → React frontend – HTTPS→ FastAPI backend (Replit deployment). On the server the CV is read (pypdf / built-in Word reader; optional Gemini for photo or complex-layout CVs) and personal data (name, email, phone, IC number, age, gender) is removed before any matching. Matched skills then pass to the three in-process models – Model 1 (AI-exposure classifier), Model 2 (arXiv monthly counts, damped trend, live refresh) and Model 3 (Catch the AI scoring) – and are fused with WEF 2025 (Malaysia), DOSM wages and MyCOL figures into an Edge Score

(low–high range, two scenarios). An ordered pathway is then built from the catalog of 53 link-checked learning resources. All catalogs are JSON files in the repository: no database, no stored user data.

Official SDC sponsor-provided GenAI API or Cloud Platform: Yes – Replit (official SDC sponsor). We used Replit to import and run the project, install dependencies, run the test suite and the course-link checker (which needs open internet), and to host the live prototype as a Replit autoscale deployment at https://sdc-2026-future-edge.replit.app. Secrets such as the optional Gemini key are kept in Replit Secrets. RunPod was not used because no part of the project needs a GPU (Participant Handbook 9.4) – all models train and run on a CPU in seconds.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we wrote five realistic personas from the problem statement – including a final-year engineering student, an admin executive with six years' experience, a call-centre team lead and a fresh CS graduate – and built synthetic CVs for each. Define: their shared question was not "which jobs match my keywords?" but "which of my skills will still be mine in three years, and what do I learn first?"

Ideate: we rejected a chatbot (it can invent jobs and courses) in favour of a forecast built from measurable signals, and replaced self-rating quizzes with a test that has known answers.

Prototype and test: every model has a held-out evaluation, 55 automated tests guard the scoring and fairness rules, and every course link was checked on 24 October 2026. A plain trend over-shot, so we switched to a damped trend; leadership was flagged "at risk", so human skills now gain AI exposure more slowly.

Deployment readiness: the prototype is live on Replit in three languages with an HR team view. Phase 2 adds a user test with XMUM students on the same pipeline.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4 – "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Target 8.5 – full and productive employment and decent work for all (supporting target).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Only 26% of Malaysian professionals who use AI at work have received formal AI training, while 56% already use AI tools (Hays Asia Salary Guide 2025, cited in the Domain 1 brief). More than 600,000 workers need reskilling within three to five years (MOSTI). The share of tasks done mainly by people is expected to fall from 47% to 33% by 2030 (WEF Future of Jobs Report 2025, Malaysia profile, p. 169). A typical career tool today gives no time-based, sourced answer: 0 of the steps in a static skills list say when a skill becomes exposed to AI.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Plan completion rate: % of test users who go from CV upload to ranked future roles and an ordered, sourced reskilling plan in 5 minutes or less. Next-step clarity: % of test users who, after using the app, can name one specific skill to learn next and a free resource for it (asked before and after use).

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	In a user test with at least 20 XMUM students and staff by the final presentation: ≥ 80% plan completion in 5 minutes or less, and an increase of at least 40 percentage points in next-step clarity (before vs after). Technical target: 100% of recommended courses link-checked within 7 days of the demo (currently 53 of 53, checked 24 Oct 2026).

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: N/A – the repository was created on 21 October 2026, after the hacking period began, and all project code was written during the hackathon.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.11+	Runtime (backend, models, data tools)	PSF License
Node.js 20+ (JavaScript ES2022)	Runtime (frontend build)	MIT
FastAPI 0.115.6	Code (REST API)	MIT
Uvicorn 0.32.1	Code (ASGI web server)	BSD-3-Clause
Pydantic 2.10.4	Code (request validation)	MIT
httpx 0.28.1	Code (HTTP client, tests)	BSD-3-Clause
python-multipart 0.0.20	Code (CV file upload)	Apache-2.0

pypdf 5.1.0 (chosen over PyMuPDF to avoid AGPL)	Code (PDF CV reading)	BSD-3-Clause
pytest 8.3.4	Code (55 automated tests)	MIT
React 18, Vite 5.4, Tailwind CSS 3	Code (frontend)	MIT
scikit-learn, pandas, matplotlib (training only, not in the running app)	Code (Model 1 training and evaluation, result charts)	BSD-3-Clause / PSF-based
GPTs-are-GPTs task exposure labels – 19,265 human-rated O*NET tasks (Eloundou et al., 2023), github.com/openai/GPTs-are-GPTs	Dataset (training labels for Model 1)	MIT
O*NET task statements and occupations, onetcenter.org	Dataset	CC BY 4.0
arXiv monthly paper counts, Oct 2023 – Sep 2026, via arXiv API (export.arxiv.org, collected 22 Oct 2026)	Dataset (trend signal for Model 2)	arXiv API Terms of Use (metadata CC0)
WEF Future of Jobs Report 2025 (incl. Malaysia profile), weforum.org	Dataset (published report, cited in the app)	WEF terms, cited by page
DOSM Salaries & Wages Survey 2025; Labour Force Survey, open.dosm.gov.my / data.gov.my	Dataset (published report)	Malaysia open data terms
TalentCorp MyCOL 2024/2025 critical occupations, talentcorp.com.my	Dataset (published report, cited in the app)	Public report, cited

Google Gemini API (default gemini-3.8-flash, configurable; Google AI Studio) – optional CV reading only	Pre-trained model (API)	Gemini API Terms of Service
Claude (Anthropic) – Claude Code / claude.ai	Pre-trained AI model (development): writing and refactoring code, tests, model cards, README, pitch-video animation and effects, original music and sound effects, and these submission documents (extent: major)	Anthropic Commercial Terms of Service
Google Gemini (chat, via Google AI Studio / gemini.com)	Pre-trained AI model (chat): drafted the first version of the 25-role dataset (titles, skills, SDG tags, descriptions) (extent: moderate)	Gemini Terms of Use
Kokoro TTS v1.0 (open-source TTS model, voice "am_michael")	Model (open source): AI-generated narrator voice for the pitch video (extent: moderate)	Apache-2.0
Replit Agent	AI development assistant: started an automatic "port to Replit" step on import; stopped after 12 seconds and all of its changes were reverted (extent: minor)	Replit Terms of Service
Hays Asia Salary Guide 2025	Dataset (published report): cited source for the Section 4 baseline – 56% of Malaysian professionals use AI tools at work; 26% have received formal AI training	Public report, cited

For completeness: the team's own datasets — the 25-role dataset (25 roles: titles, skills, SDG tags, descriptions), the skill, role, SDG and course catalogs (53 checked learning resources), and the 3 synthetic sample CVs — were created by the team (the role dataset was AI-assisted

with Gemini) and are licensed Apache-2.0 in the repository; they are listed here for transparency but are not third-party material.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns: N/A – no team member is bound by any such agreement.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Azimkhan Bekassyl

24.10.2026

